



Rossmoyne Senior High School

Semester One Examination, 2017

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One:
Calculator-free

If required by your examination administrator, please
place your student identification label in this box

Student Number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	51	35
Section Two: Calculator-assumed	13	13	100	100	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet.
3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.
4. Additional working space pages at the end of this Question/Answer booklet are for planning or continuing an answer. If you use these pages, indicate at the original answer, the page number it is planned/continued on and write the question number being planned/continued on the additional working space page.
5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
6. It is recommended that you do not use pencil, except in diagrams.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free**35% (51 Marks)**

This section has **seven (7)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1**(6 marks)**

One Australian dollar can buy 50 Indian rupees.

- (a) How many Indian rupees can be bought for 30 Australian dollars? (1 mark)
- (b) Determine how many Australian dollars are required to buy 50 000 rupees. (1 mark)
- (c) A tourist going to India wants to take 50 000 rupees with them. Their bank uses the above exchange rate but also adds a commission of 1%. How much will the rupees cost the tourist altogether? (2 marks)
- (d) Over the next few months, the exchange rate is expected to change so that one Australian dollar will buy 52 rupees. Determine the percentage increase in the exchange rate from 50 to 52 rupees. (2 marks)

Question 2**(8 marks)**

(a) Determine the value of

(i) $a^2 + 2a + 3$ when $a = 7$.

(2 marks)

(ii) $x - xy - y$ when $x = -2$ and $y = 3$.

(2 marks)

(b) Calculate the price-to-earnings ratio for a share with a market price of \$2.75 and annual earnings per share of 25 cents. **(2 marks)**

(c) The price-to-earnings ratio for a share is 20. If the market price of the share is \$1.10, determine the annual earnings per share. **(2 marks)**

Question 3**(10 marks)**

Four matrices A, B, C and D are shown below.

$$A = \begin{bmatrix} 0 & 2 & 4 \\ -3 & 1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 8 \\ 0 \\ -12 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 5 \\ -1 & 2 & 0 \\ 4 & 3 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 2 \end{bmatrix}.$$

- (a) State the size of the square matrix. (1 mark)
- (b) Calculate $2C + 5I$, where I is the 3×3 identity matrix. (3 marks)
- (c) Calculate $D \times A$. (2 marks)
- (d) Determine the values of p and q given that $p \times \begin{bmatrix} -2 \\ q \\ 3 \end{bmatrix} = B$. (2 marks)
- (e) If $E = A \times C$, determine the value of e_{21} . (2 marks)

Question 4**(6 marks)**

- (a) An investor received a dividend of \$240 on 60 shares in a mining company. Calculate the dividend per share. (1 mark)
- (b) Another company paid a dividend of 25 cents per share to an investor who held 2 000 shares in the company.
- (i) Calculate the total dividend the investor was paid. (1 mark)
- (ii) The investor had to pay income tax of 30% on the total dividend. Determine how much the investor received after tax had been deducted. (2 marks)
- (c) A shareholder has 500 shares with a market price of 40 cents in a technology company, and the company has announced a dividend per share of 1.5% of the share price. Determine the total dividend the investor can expect. (2 marks)

Question 5**(7 marks)**

A freight company has a display at a trade exhibition that includes various scale models of its standard shipping containers (such a container can be closely modelled by a cuboid). A full-size standard container is 12.0 m long, 2.4 m wide and 2.5 m high.

- (a) One of the scale models was built to a one-tenth scale of a full-size container. State the length, width and height of this model, in centimetres. (3 marks)
- (b) Another model was made to a scale of 1:100. If the area of the floor of a full-size container is 288 000 cm², determine the area of the floor of this model. (2 marks)
- (c) The largest model was built to a one-half scale of a full-size container. Determine the volume of this scale model. (2 marks)

Question 6**(6 marks)**

Ann and Ben held 400 and 350 shares respectively in three public companies: Ria, Skip and Tap.

Ann held 200 in Ria, 50 in Skip and the remainder in Tap. Ben held 50 in Tap, twice that number in Ria and the remainder in Skip.

- (a) Display this information in M , a 2×3 matrix. (2 marks)

- (b) The market value of Ria, Skip and Tap shares is \$1.20, \$3.50 and \$2.10 respectively. Display this information in matrix N , so that matrix $M \times N$ will show the values of each person's share holdings. **Do not** calculate $M \times N$. (1 mark)

- (c) Explain what information the result of $[1 \ 1] \times M$ will show. (1 mark)

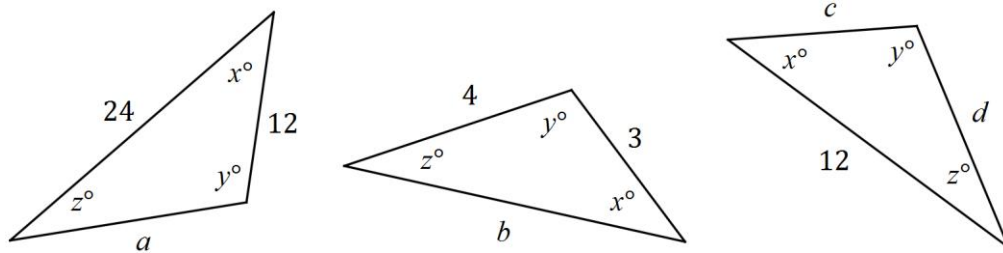
- (d) A stock market analyst expects the values of the shares in six months' time to be found from the result of $\begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 1.2 & 0 \\ 0 & 0 & 1.1 \end{bmatrix} \times N$. Use this information to calculate the expected price of Skip shares. (2 marks)

Question 7

(8 marks)

- (a) Write the scale 5 mm : 30 cm as a ratio in simplest form and without units. (2 marks)

- (b) Three similar triangles are shown below, although they are not drawn to the same scale. Some of the side lengths, in centimetres, are shown.



Determine the lengths a , b , c and d .

(4 marks)

- (c) A student used a map drawn to a scale of 1: 10 000 to plan a walk. The length of the walk on the map was 5.5 cm. State, with reasoning, whether the length of the walk was more than a kilometre. (2 marks)

Additional working space

Question number: _____

Additional working space

Question number: _____

